

25/09/26

git clone "repo url"

ls

git clone "repo url" new folder => new folder name

cd newfolder

ls

2 way to open the file

explorer => windows -> file -> open -> Pandrathur

start file name => window shell -> file open again.

history

git branch -M master

ls

git status

cd ..

git status

git add new folder

git checkout main => branch to branch switch

branch to branch switch pannalum any one branch -> git add pannitu another branch -> check pannalum show pannum.

git commit -m "

Commit the main branch.

staging area - la endha branch - la commit pannamo
only that branch - la mattum tha kaatum.

working directory to staging area using git
add ~~to~~ one branch it will show the
~~another~~ branch.

master branch - la $\Rightarrow$ test file untracked

main branch $\Rightarrow$ test untracked

main branch - la test commit,
after go to the master.

master branch $\Rightarrow$ test commit pannatha
kaatum.

master $\rightarrow$ commit test.

go to the main

git status $\Rightarrow$ ^{it will not} ~~don't~~ show the ^{test} file.

because master branch - la save aadukche.

Git merge:

git merge master

ls

git checkout master

ls

ReBase:

which commit is merged that is added.

git rebase main

↳ main la inukuna files ellam master-ku vandhedum.

git rm rebase file

ls

git commit -m "deleted rebase file" ⇒ permanently.

git push origin main

↳ remote branch.

stash ⇒ Temporary save.
Master branch

git add stash

git status ⇒ working to stash area.

git stash ⇒ store the file inside stash

ls

main branch:

git stash pop

ls

git status

Instance ⇒ virtual physical server replica.

hypervisor ⇒ virtualization helper tool.

↳
Containerization ⇒ virtualize the OS.

Virtualization ⇒ virtualize the hardware.

Virtualization $\Rightarrow$ replace the hardware with help of hypervisor. vmware, virtual box.

Containerization $\Rightarrow$ virtualize the OS,
tool $\Rightarrow$ Docker, podman, containersd

task:

create VM

name $\Rightarrow$ ubuntu server.

using OST $\Rightarrow$ ubuntu server OS.

Monolithic $\Rightarrow$ any one affect it leads to affect the whole server.

Microservice $\Rightarrow$ separate containers
so, any one affect it is not leads to affect the whole because it separately each one.

Containers like additional layer on OS.

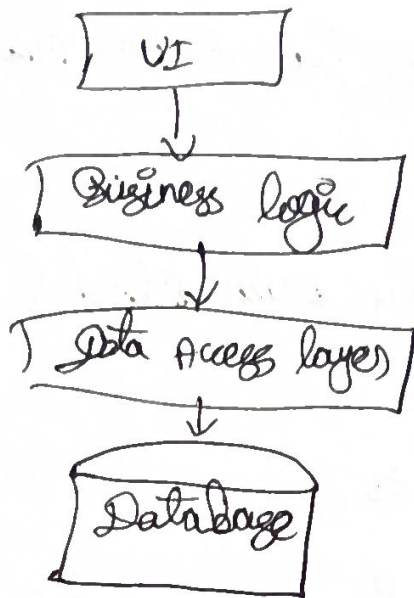
Microservice $\Rightarrow$ no maintenance

Docker Registry $\Rightarrow$ storage component for docker images.

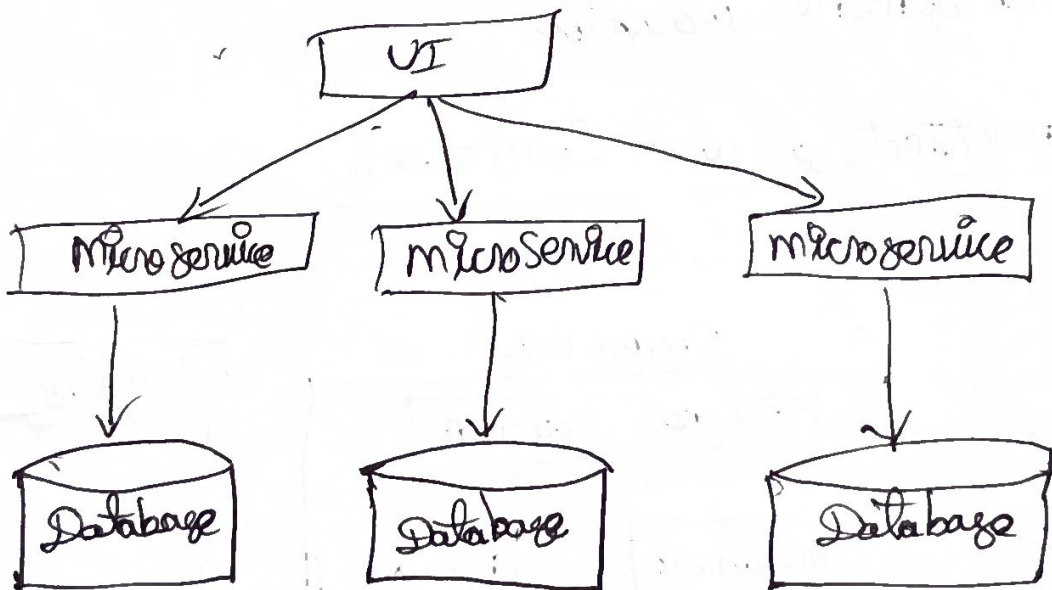
Docker hub $\Rightarrow$ the images stored in docker hub.

Docker Daemon $\Rightarrow$ backend $\Rightarrow$ ~~back~~ Docker Engine.

Monolithic Architecture



Microservice Architecture



Micro service:

The idea behind microservices is that some types of applications become easier to build and maintain when they are broken down into smaller, composable pieces which work together. Each component is developed separately, and the application is then simply the sum of its constituent components.

Docker Registry:

Docker Registry is a storage component for Docker Images.

we can store the images in either public / Private repositories.

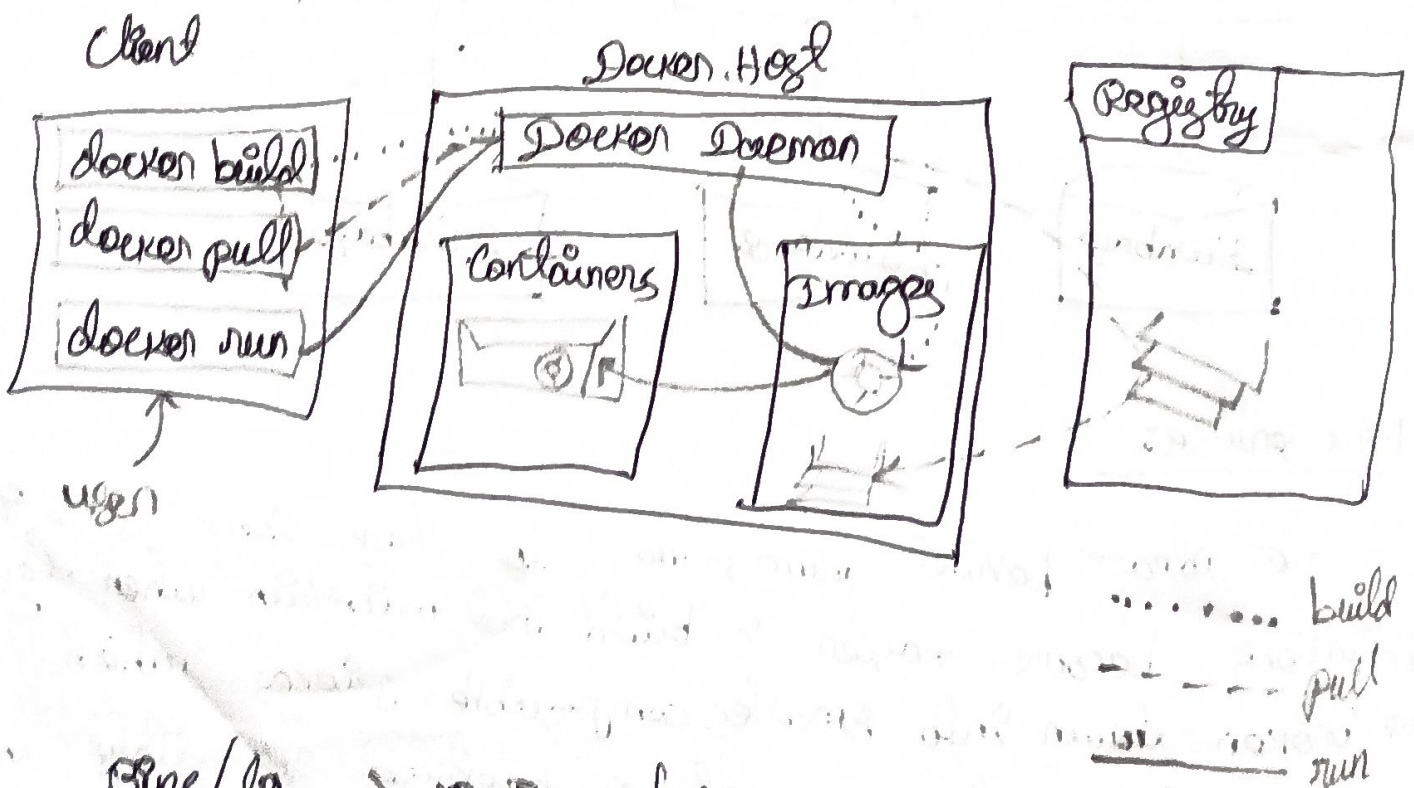
Docker hub is Docker's very own cloud repository.

Why use docker registries?

* Control where your images are being stored.

* Integrate image storage with your in-house development workflow.

Docker Architecture and Components:



Bins/libs $\Rightarrow$ Binaries / Libraries

Container Layer

$\rightarrow$ Read & write

Layer 1 - RUN yum install $\rightarrow$ wget

$\rightarrow$ Read only

Base Layer - FROM centos:7

$\rightarrow$ Read only

WSL $\Rightarrow$ windows subsystem layer.

1. docker -v

2. docker search node

image $\Rightarrow$ alpine/node
 $\searrow$ uploader name
 (or) user name.

3. docker pull image name $\Rightarrow$ docker pull node.
 $\searrow$ local-km image download pandathu
 local repo - la inukura images left

4. docker images

Content Size $\Rightarrow$ image uI size.

web server host panna use pandra Engine $\Rightarrow$ Apache
Apache $\Rightarrow$ used. when light weight webpage hosting.
 (or) glu website hosting.

Redhat $\Rightarrow$ Apache name httpd
docker search apache2

docker ~~image~~ search httpd

docker pull httpd

docker images

2 Commands

1. Create $\Rightarrow$ only container create

2. RUN $\Rightarrow$ It create container after start (i.e) power on.

docker create httpd $\Rightarrow$ create image.

docker ps -a $\Rightarrow$ ^{↳ apache $\Rightarrow$ centos} It shows the created containers

docker start container name (or) container-id:

↓
reply in

container name must be show.

docker ps -a $\Rightarrow$ it shows the all containers names.

apache default port $\Rightarrow$ 80/tcp.

RDP port $\Rightarrow$ 3389

containers $\Rightarrow$ running containers only show

↓
command $\Rightarrow$ docker ps.

docker create --name demoname httpd
↳ what name ↳ which image

docker ps

docker start demoname

docker images

docker pull httpd:2.4.68

(i.e.) httpd:latest

↳ current version.

docker create --name demoname httpd:2.4.68

if not mention $\Rightarrow$ automatically create latest version.